Exercises

Algebraic problems

Question 6.1: Find the following indefinite integrals:

(a) $\int x^2 dx$
(b) $\int 12x^4 dx$
(c) $\int sin(x) dx$
(d) $\int cos(x) dx$
(e) $\int e^x dx$
(f) $\int e^{2x} dx$
(g) $\int x^2 + 2x + 3 dx$
(h) $\int sin(x) + cos(x) dx$
(i) $\int e^{2x} + 2x dx$
(j) $\int e^x + cos(x) dx$

Question 6.2: Find the integrals between the limits specified:

(a) $\int_0^{10} x^2 dx$
(b) $\int_{-10}^{10} x^2 dx$
(c) $\int_{-1}^{1} x^3 dx$
(d) $\int_0^{80} \left(-0.1x^2 + 10x\right) dx$
(e) $\int_0^{50} \left(-0.1x^2 + 10x\right) dx$
(f) $\int_0^1 \exp(3x)\, dx$

Question 6.3: Find the integrals between the limits specified:

(a) $\int_0^b \frac{h}{b} x dx$ (i.e. the area of a triangle with base b and height h)
(b) $\int_0^b h dx$
(c) $\int_0^T \exp(-\lambda t)\, dt$

Question 6.4: Use the fact that $\int ax^n = \frac{ax^{n+1}}{n+1} + d$ to integrate the following:

(a) $f(x) = \sqrt{x}$
(b) $f(x) = \frac{1}{x^2}$
(c) $f(x) = \frac{1}{\sqrt{x}}$

Question 6.5: Use separation of variables to integrate the following ODE:

$$\frac{dN}{dx} = 1/N$$

Real-world problems

Question 6.6: Water flows down a river at a rate $\frac{dW}{dt}$ (m^3 s^{-1}) described by:

$$\frac{dW}{dt} = 1000t^{0.5}$$

Determine how much water (m^3) flows down the river between $t = 100$ and $t = 700$ s?

Question 6.7: The rate (in mm/hr) that rain falls from a cloud between time $t = 3.5$ and $t = 6.5$ hours can be approximated by $-3t^2 + 30t - 65$. Calculate the total precipitation, in mm, that falls between $t = 3.5$ and $t = 6.5$ hours.

Question 6.8: The radioactive decay equation. Solve the radioactive decay equation by separation of variables and integration. Recall that it states that the number of decays per unit time is proportional to the number of unstable nuclei:

$$\frac{dN}{dt} = -\lambda N$$

where N is the number of unstable nuclei, t is time and λ is called the decay constant, $\lambda = \frac{\ln 2}{t_{1/2}}$ where $t_{1/2}$ is the half-life of the sample.

Question 6.9: Imagine a sheep was allowed to graze in a field that had been contaminated by ^{137}Cs released at Chernobyl. About a kilogram of ^{137}Cs was released, but only a small fraction of this was atomised and subsequently spread over a very wide area. The sheep had been eating grass that contained a surface density of radioactive nuclei. That density changes over time as the radioactive material decays. If we assume that the the surface density of atoms starts at a value of $2.16 \times 10^9 m^{-2}$, and the surface area A is 7.47 m^2, integrate the following equation to calculate the number of atoms ingested over 1 year:

$$A \int_{t=0}^{t=1} N(t)dt$$

where $N(t)$ is the radioactive decay equation. Note that, ^{137}Cs is a beta emitter with a half life of 30 years (it decays to ^{137}Ba).

Question 6.10: The hydrostatic equation. Solve the hydrostatic equation by separation of variables and integration. Recall that it states the density of air at altitude z depends on the pressure, P, and temperature of the air, T, (through the ideal gas law): $\rho = \frac{P}{R_a T}$, where $R_a = 287$ J kg^{-1} K^{-1}:

$$\frac{dP}{dz} = -\frac{P}{R_a T} g$$